

Tutorial 2 - Answers

Question 1

Proof. For $n \geq 1$, the binomial theorem gives us:

$$(1+x)^n = \sum_{k \in \mathbb{N}} \binom{n}{k} x^k$$

This sum is finite as per the hint; terms where $k > n$ equal 0. To get alternating signs, substitute $x = -1$:

$$0 = (1 + (-1))^n = \sum_{k \in \mathbb{N}} \binom{n}{k} (-1)^k$$

We can split the sum into two pieces: where k is even, and one where k is odd.

$$0 = \sum_{k \in \mathbb{N}} \binom{n}{2k} (-1)^{2k} + \sum_{k \in \mathbb{N}} \binom{n}{2k+1} (-1)^{2k+1}$$

Since $2k$ is always even, $(-1)^{2k} = 1$. Since $2k+1$ is always odd, $(-1)^{2k+1} = -1$:

$$\begin{aligned} 0 &= \sum_{k \in \mathbb{N}} \binom{n}{2k} (1) + \sum_{k \in \mathbb{N}} \binom{n}{2k+1} (-1) \\ &= \sum_{k \in \mathbb{N}} \binom{n}{2k} - \sum_{k \in \mathbb{N}} \binom{n}{2k+1} \end{aligned}$$

Rearranging both sides yields:

$$\sum_{k \in \mathbb{N}} \binom{n}{2k} = \sum_{k \in \mathbb{N}} \binom{n}{2k+1}$$

■

Question 2

Proof. We proceed by induction. Let $P(n)$ be the statement that $\sum_{i=0}^n \binom{n}{i} 9^i = 10^n$ for all $n \in \mathbb{N}$.

Base Case ($n = 0$):

$$10^0 = 1 = \sum_{i=0}^0 \binom{0}{i} 9^i$$

Since LHS = RHS, the base case $n = 0$ holds true.

Inductive Hypothesis. Assume that $P(k)$ is true for some integer $k \geq 0$, meaning:

$$\sum_{i=0}^k \binom{k}{i} 9^i = 10^k$$

Inductive Step. We want to show that $P(k+1)$ is true, meaning we want to evaluate:

$$\sum_{i=0}^{k+1} \binom{k+1}{i} 9^i$$

Using **Pascal's Identity** for binomial coefficients ($\binom{k+1}{i} = \binom{k}{i} + \binom{k}{i-1}$), substitute this into the summation:

$$= \sum_{i=0}^{k+1} \left[\binom{k}{i} + \binom{k}{i-1} \right] 9^i$$

Split this into two separate sums:

$$= \sum_{i=0}^{k+1} \binom{k}{i} 9^i + \sum_{i=0}^{k+1} \binom{k}{i-1} 9^i$$

Extract the zero terms ($\binom{k}{k+1} = 0$ and $\binom{k}{-1} = 0$) to align the bounds:

$$= \left(\sum_{i=0}^k \binom{k}{i} 9^i \right) + \left(\sum_{i=1}^{k+1} \binom{k}{i-1} 9^i \right)$$

In the second sum, perform a index shift by letting $j = i - 1$ (so when $i = 1$, $j = 0$; and when $i = k + 1$, $j = k$):

$$= \left(\sum_{i=0}^k \binom{k}{i} 9^i \right) + \left(\sum_{j=0}^k \binom{k}{j} 9^{j+1} \right)$$

Factor out a 9 from the second sum:

$$= \sum_{i=0}^k \binom{k}{i} 9^i + 9 \left(\sum_{j=0}^k \binom{k}{j} 9^j \right)$$

By our **Inductive Hypothesis**, both summations equal 10^k :

$$= 10^k + 9 \cdot 10^k$$

Factor out 10^k :

$$= (1 + 9) \cdot 10^k$$

$$= 10 \cdot 10^k$$

$$= 10^{k+1}$$

This completes the inductive step. ■

Question 3

Here is a heuristic argument before we proceed by induction. Taking a few derivatives of $f(x)$ yields the following.

First derivative $k = 1$:

$$\frac{d}{dx}(\ln(x)) = -\frac{1}{x}$$

Second derivative $k = 2$:

$$\frac{d^2}{dx^2} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

Third derivative $k = 3$:

$$\frac{d^3}{dx^3} \left(-\frac{1}{x^2} \right) = \frac{2}{x^3}$$

Fourth derivative $k = 4$:

$$\frac{d^4}{dx^4} \left(\frac{2}{x^3} \right) = -\frac{6}{x^4}$$

Observing the pattern, we can notice a few things namely:

1. The denominator is always x^n , where n is the derivative's order.
2. The numerator is precisely $(n-1)!$.
3. For odd k , $\frac{d^n}{dx^n} > 0$ while for even k , $\frac{d^n}{dx^n} < 0$.

Combining these patterns into one formula, we conjecture that the formula of the n -th derivative of $f(x)$ is given by:

$$\frac{d^n}{dx^n} = (-1)^{n-1} \frac{(n-1)!}{x^n}$$

Proof. We now proceed by induction. Let

$$P(n) : \frac{d^n}{dx^n}(\ln(x)) = (-1)^{n-1} \frac{(n-1)!}{x^n}$$

for all $n \geq 1, n \in \mathbb{Z}$.

Base case ($n = 1$). Check if the left hand side equals the right.

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x} = (-1)^{1-1} \frac{(1-1)!}{x^1} = \dots = \frac{1}{x}$$

The base case $P(1)$ is true.

Inductive step. Assume $P(n)$ holds for some arbitrary $k \geq 1, k \in \mathbb{Z}$. That is, assume the inductive hypothesis:

$$\frac{d^k}{dx^k}(\ln(x)) = (-1)^{k-1} \frac{(k-1)!}{x^k} \quad (1)$$

We need to show that $P(k) \implies P(k+1)$:

$$\frac{d^{k+1}}{dx^{k+1}}(\ln(x)) = (-1)^k \frac{k!}{x^{k+1}}$$

Apply the operator $\frac{d}{dx}$ to the k -th derivative:

$$\frac{d^{k+1}}{dx^{k+1}}(\ln(x)) = \frac{d}{dx} \left(\frac{d^k}{dx^k}(\ln(x)) \right) \quad (2)$$

Substituting (1) into (2):

$$\begin{aligned} \frac{d^{k+1}}{dx^{k+1}}(\ln(x)) &= \frac{d}{dx} \left[(-1)^{k-1} \frac{(k-1)!}{x^k} \right] \\ &= \frac{d}{dx} [(-1)^{k-1} (k-1)! x^{-k}] \end{aligned}$$

Pulling out constants and by the power rule:

$$\begin{aligned} &= (-1)^{k-1} (k-1)! \cdot \frac{d}{dx}(x^{-k}) \\ &= (-1)^{k-1} (k-1)! \cdot (-k) x^{-k-1} \\ &= (-1)^{k-1} (k-1)! \cdot (-1) [(k) x^{-k-1}] \\ &= (-1)^k k! x^{-k-1} \end{aligned}$$

We can rewrite this into a more familiar way:

$$(-1)^k \frac{k!}{x^{k+1}}$$

Which is precisely $\frac{d^{k+1}}{dx^{k+1}}(\ln(x))$. Therefore, by induction, $P(n)$ holds true. ■

Question 4

Proof. Let $k \in \mathbb{N}$ represent the number of positive integers in the sum. We are looking for the number of integer solutions to $x_1 + x_2 + \cdots + x_k = n$, where $x_i \geq 1$ and order matters.

Distributing n indistinguishable units into k distinguishable, non-empty bins is equivalent to choosing $k - 1$ dividers from the $n - 1$ spaces between the units. Thus, for a fixed k , the number of ways is:

$$\binom{n-1}{k-1}$$

The number of parts k can range from 1 to n . The total number of ways can be written as a sum:

$$\sum_{k=1}^n \binom{n-1}{k-1}$$

Let $j = k - 1$. As k ranges from 1 to n , j ranges from 0 to $n - 1$:

$$\sum_{j=0}^{n-1} \binom{n-1}{j}$$

By the Binomial Theorem, the sum of all binomial coefficients for a given power is:

$$\sum_{j=0}^{n-1} \binom{n-1}{j} (1)^{n-1-j} (1)^j = (1+1)^{n-1} = 2^{n-1}$$

Accounting for order, the number of ways n can be written as the sum of strictly positive integers is 2^{n-1} . ■

Question 5

Proof (Strong Induction). Let

$$P(n) : x^n + \frac{1}{x^n} \in \mathbb{N}, n \in \mathbb{N}.$$

Base cases. Let $n = 0$:

$$x^0 + \frac{1}{x^0} = 2 \in \mathbb{N}$$

Now let $n = 1$:

$$x^1 + \frac{1}{x^1} = x + \frac{1}{x} \in \mathbb{N}$$

which is precisely $P(n)$ again.

Inductive step. Assume $P(m)$ is true for all $0 \leq m \leq k$, where $k \geq 1$. Show that $P(k+1)$ is true, that is:

$$x^{k+1} + \frac{1}{x^{k+1}} \in \mathbb{N}.$$

We can expand the product of the terms for $n = 1$ and $n = k$:

$$\left(x + \frac{1}{x}\right) + \left(x^k + \frac{1}{x^k}\right) = x^{k+1} + x^{k-1} + \frac{1}{x^{k-1}} + \frac{1}{x^{k+1}}$$

We can rearrange and isolate $x^{k+1} + \frac{1}{x^{k+1}}$:

$$\left(x + \frac{1}{x}\right) \left(x^k + \frac{1}{x^k}\right) - \left(x^{k-1} + \frac{1}{x^{k-1}}\right) = x^{k+1} + \frac{1}{x^{k+1}} \quad (1)$$

Because $k \geq 1$, we have $0 \leq k-1 \leq k$. By the inductive hypothesis, all terms in the left hand side of (1) are all in $\mathbb{N}$. Therefore, the sum is in $\mathbb{N}$. ■

Proof (Binomial Theorem). Let

$$x + \frac{1}{x} = k, k \in \mathbb{N}.$$

Using the binomial theorem, the n -th power of this expansion is

$$k^n = \left(x + \frac{1}{x}\right)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} \left(\frac{1}{x}\right)^j$$

Simplifying using the law of exponents ($x^{n-j} \cdot x^{-j} = x^{n-2j}$):

$$k^n = \sum_{j=0}^n \binom{n}{j} x^{n-2j}$$

Isolating $j = 0$ and $j = n$:

$$k^n = \left(\binom{n}{0} x^n + \binom{n}{n} \frac{1}{x^n}\right) + \sum_{j=1}^{n-1} \binom{n}{j} x^{n-2j}$$

Rearrange to explicitly solve $x^n + \frac{1}{x^n}$:

$$x^n + \frac{1}{x^n} = k^n - \sum_{j=1}^{n-1} \binom{n}{j} x^{n-2j}$$

Because $\binom{n}{j} = \binom{n}{n-j}$ and the powers x^{n-2j} and $x^{-(n-2j)}$ are reciprocals, the terms inside the sum from $j = 1$ to $n - 1$ are symmetric. Rewrite the sum by pairing each term with its corresponding reciprocal counterpart:

If n is odd, the sum has an even number of terms and pairs up completely:

$$\sum_{j=1}^{n-1} \binom{n}{j} x^{n-2j} = \sum_{j=1}^{(n-1)/2} \binom{n}{j} \left(x^{n-2j} + \frac{1}{x^{n-2j}} \right)$$

If n is even there is a middle term when $j = \frac{n}{2}$ (which equals just the constant binomial coefficient) while the remaining terms pair up:

$$\sum_{j=1}^{n-1} x^{n-2j} = \binom{n}{n/2} + \sum_{j=1}^{n/2-1} \binom{n}{j} \left(x^{n-2j} + \frac{1}{x^{n-2j}} \right)$$

$k^n \in \mathbb{N}$ (as $k \in \mathbb{N}$ and $\mathbb{N}$ is closed under powers). By recursive application of the binomial expansion on lower powers, each symmetric pair evaluates to an integer polynomial in terms of k . The binomial coefficients $\binom{n}{j}$ are integers as well. Thus, $\left(x^{n-2j} + \frac{1}{x^{n-2j}} \right)$ is expressed as a natural number minus a combination of integer-valued products, which evaluates cleanly to a natural number. ■